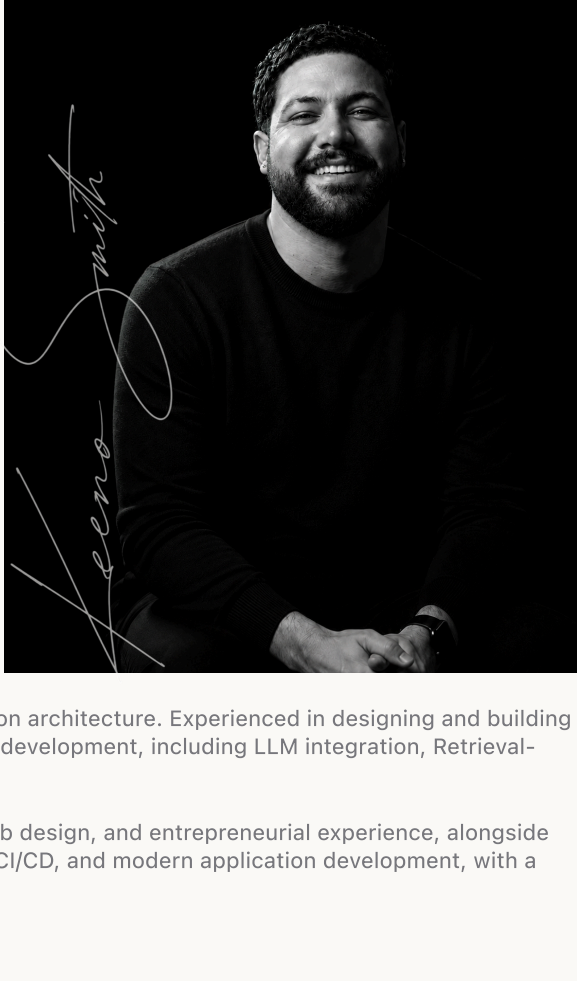


github.com/keenosmith-del
linkedin.com/in/keenotraysmith
keenosmith.vercel.app/

business.keenosmith@icloud.com
+27 82 415 8441



Keeno Smith

Full-Stack Software Engineer & Web Developer

AI Application Developer · Generative & Agentic AI · Cloud & DevOps

Full-Stack Software Engineer & Web Developer with a strong foundation in JavaScript, React, Node.js, and modern web application architecture. Experienced in designing and building end-to-end software across frontend, backend, APIs, databases, and deployment, with a growing specialization in AI application development, including LLM integration, Retrieval-Augmented Generation (RAG), prompt engineering, agentic AI, and AI-driven workflows.

My background combines formal software engineering and computer science education with hands-on product development, web design, and entrepreneurial experience, alongside growing expertise across the AWS, Microsoft Azure, and Google Cloud ecosystems. I work across cloud infrastructure, DevOps, CI/CD, and modern application development, with a focus on building scalable, production-oriented software and integrating emerging technologies into practical applications.

01 Education

- Generative AI**
2026 · Stellenbosch University
Generative AI training focused on modern AI concepts, generative models, AI application development, and practical applications of emerging AI technologies.
 - Full-Stack Software Engineering & Web Development**
2026 · Hyperion Development
Full-stack software engineering and web development training covering frontend, backend, databases, APIs, software architecture, and application deployment.
 - BSc Computer Science**
2017 - Present (Coursework Completed) · University of South Africa
Computer Science degree studies covering core programming, software development, computer science, and related technical disciplines.
 - National Senior Certificate**
2015 · Greenside High School
- Microsoft Applied Skills: Microsoft Defender XDR**
2026 · Microsoft
Demonstrating practical security operations skills with Microsoft Defender XDR, including incident management, endpoint investigation, threat detection and Advanced Hunting with KQL.
 - Microsoft Agent Architect
Cyber Security & Threat Defense**
2026 · Microsoft x Founderz
Architect-level training focused on designing scalable AI agent solutions using Microsoft Foundry, including agent architecture, project-based implementation, security labs, simulations, prototyping, and production-oriented approaches to enterprise AI.

02 Certifications

AWS Skill Builder: [skillsprofile.skillbuilder.aws/user/keenosmith](#)
Microsoft Learn: [learn.microsoft.com/en-us/users/keenosmith/](#)
Credly: [credly.com/users/keeno-smith](#)

- AWS AI / ML Scholars Programme AI Practitioner**
- AWS AI Programmer Nanodegree**
2026 · AWS x Udacity
Project-based AI/ML development covering Python, NumPy, Pandas, neural networks, PyTorch, transformer architectures, generative AI, prompt engineering, and AWS AI/ML services.
- Generative AI for Software Engineers**
- AI Course for Developers**
2025 / 2026 · WeThinkCode_...
Applied generative AI development covering LLM integration, prompt engineering, AI-assisted software development, agentic workflows, API integration, and incorporating generative AI capabilities into software applications.

03 Experience

- Independent Software Engineering & Web Development**
2022-2026 Project-Based · Freelance & Independent Development
Software engineering and web development experience built through freelance, independent, and project-based development, with a focus on designing and delivering end-to-end applications. Work spans full-stack engineering, REST APIs, database architecture, authentication, cloud deployment, AI application development, and modern web architecture, taking applications from requirements and interface design through implementation, integration, testing, and production delivery.

Full-Stack Software Engineer
Project: Authenticated MERN Productivity Platform
Designed and developed an end-to-end productivity platform using React, Node.js / Express, and MongoDB, with JWT authentication, interconnected resources, account lifecycle management, and cloud deployment. The project covers application architecture, secure API design, data modelling, state management, and production delivery across the frontend and backend.
Skills Demonstrated
Application architecture, React development, REST API development, database design and data modelling, authentication and authorization, JWT security, password security, asynchronous programming, client-server state management, API integration, file processing, account lifecycle management, environment configuration, Git-based development, and cloud deployment.
Technology Stack
React, Vite, JavaScript, React Router, Tailwind CSS, Node.js, Express, MongoDB, Mongoose, JWT, bcrypt, Multer, REST APIs, Fetch API, Git, GitHub, and cloud deployment platforms.
Responsibilities
 - Designed the application architecture across the React frontend, service layer, Express REST API, authentication middleware, Mongoose models, and MongoDB persistence layer.
 - Implemented secure authentication workflows using JWT, Bearer-token authorization, protected routes, bcrypt password hashing, and credential verification.
 - Developed authenticated REST API services and client-side service modules supporting resource CRUD, account management, data export, and workspace lifecycle operations.
 - Modelled interconnected MongoDB data with Mongoose, including ownership boundaries, Objectid relationships, schema definitions, validation, and cross-resource references.
 - Implemented centralized authentication state, asynchronous resource orchestration, client-side state reconciliation, controlled forms, validation, and authenticated client-server communication.
 - Developed account and workspace lifecycle functionality including profile management, password operations, data export, workspace clearing, account deletion, and multipart profile asset processing.
 - Configured environment-specific endpoints and production builds, maintained source control through Git and GitHub, and deployed the frontend and backend across cloud-hosted environments.
- Software Development Engineer**
Project: API-Driven Full-Stack Web Application
Developed an API-driven full-stack application integrating a React frontend, Node.js / Express REST API, MongoDB persistence, and multiple external data providers. Implemented asynchronous API workflows, response normalisation, persistent user libraries, search, application state management, and browser-based media handling.
Skills Demonstrated
REST API development, API integration, backend architecture, data modelling, MongoDB persistence, asynchronous programming, client-server state management, data normalisation, browser APIs, and Git-based development.
Technology Stack
React, JavaScript, Vite, React Router, Node.js, Express, Mongoose, Axios, Fetch API, Deezer API, iTunes Search API, RSS Parser, Git, GitHub, and Oxlint.
- Full-Stack Software Engineer / AI Application Engineer**
Project: Retrieval-Augmented Generative AI Knowledge Engineering Platform
Designed and developed a full-stack SaaS AI platform integrating LLM inference, Retrieval-Augmented Generation, semantic embeddings, vector retrieval, persistent knowledge, configurable AI context, document processing, and streamed model responses. The platform combines AI application engineering with knowledge engineering, API development, data modelling, and modular software architecture.
Skills Demonstrated
Generative AI application development, LLM integration, RAG, knowledge engineering, semantic embeddings, vector retrieval, cosine similarity, Top-K ranking, prompt engineering, context engineering, document processing, text chunking, streaming architectures, REST API development, SaaS architecture, database modelling, asynchronous programming, API security, validation, QA, and technical product development.
Technology Stack
React, JavaScript, Vite, Node.js, Express, MongoDB, Mongoose, Groq SDK, Groq LLM inference, OpenAI-compatible model architectures, Hugging Face Transformers, DNNX, all-MiniLLM-L6-v2, cosine similarity, RAG, Multer, Mammoth, PDF Parse, CORS, Fetch API, Server-Sent Events, environment configuration, Oxlint, Git, and GitHub.
Responsibilities
 - Designed the platform architecture across the React client, Express API, MongoDB persistence, document-processing pipeline, embedding layer, retrieval services, and LLM inference layer.
 - Engineered RAG workflows covering document ingestion, overlapping text chunking, semantic embedding generation, vector persistence, cosine similarity calculation, similarity ranking, and Top-K retrieval.
 - Implemented local semantic embedding infrastructure using Hugging Face Transformers, ONNX, and all-MiniLLM-L6-v2, including lazy model-loading, mean pooling, vector normalisation, and embedding generation for knowledge chunks and search queries.
 - Developed document processing for TXT, Markdown, PDF, and DOCX content using Multer, PDF Parse, and Mammoth, including multipart uploads, content extraction, validation, normalisation, chunking, and persistence.
 - Integrated server-side LLM inference through the Groq SDK and implemented streamed responses using asynchronous provider streams and Server-Sent Events, with incremental client-side parsing, completion handling, error processing, and request cancellation.
 - Designed configurable AI context workflows incorporating conversation history, memory, knowledge, personality, custom instructions, model selection, temperature, and response-token configuration, with prompt engineering separating user input, context, retrieval results, and model inference.
 - Designed and maintained MongoDB/Mongoose models and REST API workflows for conversations, messages, memories, knowledge documents, embedded vectors, semantic search, document ingestion, and streamed AI inference.
 - Implemented application security and reliability controls through environment-based configuration, server-side AI credentials, CORS policy, request validation, Mongoose schema validation, controlled document uploads, build validation, linting, and runtime error handling.
- Full-Stack Software Engineer / Database Application Engineer**
Project: Enterprise Database Workspace & Runtime Schema Management Platform
Designed and developed a full-stack database workspace centred on PostgreSQL, SQL, relational modelling, Prisma ORM, and runtime schema management. The platform provides browser-based management of tables, columns, constraints, relationships, records, and SQL queries, combining ORM-based data access with direct PostgreSQL SQL and runtime DDL operations.
Technology Stack
PostgreSQL, SQL, Prisma ORM, Prisma CLI, Prisma Migrations, Runtime DDL, information_schema, PostgreSQL system catalogs, Node.js, Express, REST APIs, React, JavaScript, and Fetch API.
Responsibilities
 - Designed the PostgreSQL architecture and relational models, including primary and foreign keys, unique constraints, nullability, defaults, identity columns, and relationship integrity.
 - Engineered runtime schema management for dynamically creating tables and adding, modifying, renaming, and deleting columns and constraints through PostgreSQL DDL.
 - Developed a dedicated SQL builder and validation layer for DDL generation, including identifier quoting and escaping, data-type allowlisting, type normalisation, and constraint validation.
 - Implemented PostgreSQL schema introspection using `information_schema` and PostgreSQL system catalogs to reconstruct tables, columns, constraints, relationships, defaults, and identity properties.
 - Implemented transactional schema operations and metadata synchronisation between application-level workspace definitions and physical PostgreSQL structures, while integrating Prisma ORM, CLI, and migrations for structured database operations.
 - Developed Express REST APIs for relational CRUD, dynamic record operations, schema management, relationship operations, database health monitoring, and SQL execution through `POST /api/query`, with PostgreSQL-aware integrity validation, conflict handling, state verification, and repeatable database seeding.
- Co-Founder & Brand Strategist**
2023-2026 · Firstdaes Entertainment
Co-founded and helped lead an events business, overseeing brand strategy, market positioning, audience development, and promotional operations. Managed end-to-end marketing initiatives and campaigns, translating business objectives into customer-facing strategies and coordinating ad projects from initial concept through execution. Contributed to commercial planning, operational decision-making, and the development of the company's broader brand and growth strategy.
- Technical Contract Specialist**
2019-2022 · 3Play Media
Reviewed, transcribed, proofread, and edited technical documentation, training materials, and technology-focused content across enterprise software, cloud computing, networking, cybersecurity, software development, and emerging technologies. Worked with technically complex material requiring accuracy, terminology consistency, contextual understanding, and close attention to technical detail. Content covered technologies and subject matter associated with Cisco, Oracle, Palo Alto Networks, Unreal Engine, Deloitte, ThousandEyes, Salesforce, NVIDIA, and Apple.

04 Cloud, DevOps & Platform Engineering

- AWS**
Hands-on cloud development and applied AI/ML learning across AWS infrastructure and services including EC2, S3, IAM, VPC, RDS, ECS, EKS, Kubernetes, SageMaker, Bedrock, and related developer and deployment tooling, supported by practical labs, simulated cloud environments, and project-based learning through AWS Skill Builder.
AWS Skill Builder: [skillsprofile.skillbuilder.aws/user/keenosmith](#)
- Microsoft**
Hands-on development and applied cloud learning across Microsoft Azure and its AI ecosystem, including Azure App Service, Azure Functions, Storage, Azure SQL, Cosmos DB, AKS, Azure Container Registry, Microsoft Entra ID, Azure Monitor, Microsoft Foundry, Azure AI services, and enterprise security tooling such as Microsoft Defender XDR, supported by practical labs, simulations, and project-based environments through Microsoft Learn.
Microsoft Learn: [learn.microsoft.com/en-us/users/keenosmith/](#)
- Google Cloud**
Hands-on cloud and AI/ML development across Google Cloud services including Compute Engine, Cloud Storage, Cloud Run, GKE, Kubernetes, Cloud Functions, Cloud SQL, Firestore, BigQuery, Vertex AI, Gemini, Artifact Registry, Cloud Build, and Google Cloud IAM, supported by practical labs, simulated environments, developer resources, and structured cloud learning through Google Skills and Google Developers.
Google Skills: [skills.google/public_profiles/105079cb-27bf-46a4-9c48-9ce8fc594527](#)
Google Developer Group: [g.dev/keenosmith](#)

05 Projects

Portfolio: [keenosmith.vercel.app/](#)
GitHub: [github.com/keenosmith-del](#)

Software Engineering

Full-Stack Development
Frontend Development
Backend Development
API Development
Application Architecture
Component Architecture
State Management

AI & Agentic Systems

AI Application Development
Generative AI
Agentic AI
Multi-Agent Systems
LLM Integration
RAG & Knowledge Retrieval
Tool Calling
AI Evaluation

Data & Knowledge Engineering

MongoDB & Mongoose
PostgreSQL & SQL
Prisma ORM
Database Design
Data Modelling
Semantic Embeddings
Vector Retrieval

Cloud, DevOps & Delivery

Cloud Deployment
CI/CD
Git & GitHub
Docker
Kubernetes
Authentication & Authorization
Observability
Testing & Evaluation

- Multi-Agent AI Systems - Microsoft**
Agentic AI · Multi-Agent Architecture
Microsoft Foundry · OpenAI Models · RAG · Tool Calling
Multi-agent AI systems developed with Microsoft Foundry and OpenAI models, using specialised agents, agent orchestration, RAG, tool calling, structured outputs, evaluation, observability, scoped permissions, and human-in-the-loop workflows across software engineering, DevOps, insurance, and IT service management.

AI-Powered Software Engineering & DevOps Operations Platform
Multi-agent engineering and operations platform separating software analysis, incident investigation, and DevOps resolution into specialised agents. Integrates repository code inspection, pull requests, documentation, logs, monitoring data, deployment history, RAG, GitHub workflows, remediation recommendations, and human approval.
Microsoft Foundry · GitHub · RAG · Observability · CI/CD
GitHub Repo: [github.com/keenosmith-del/ai-powered-devops-incident-intelligence](#)

AI-Powered Insurance Claims Processing
Multi-agent claims processing architecture separating document intake, policy and evidence assessment, and fraud and risk analysis. Uses document processing, knowledge retrieval, structured outputs, business rules, risk scoring, evaluation datasets, observability, auditability, responsible AI controls, and human approval for governed decision workflows.
Microsoft Foundry · Multi-Agent Architecture · RAG · Evaluation
GitHub Repo: -

AI-Powered IT Service Desk & Incident Management
Multi-agent incident management architecture separating incident triage, technical investigation and workflow orchestration. Uses historical incidents, service knowledge, monitoring data, application logs, deployment history, repositories, diagnostic APIs, RAG, structured incident assessment, root-cause analysis, resolution recommendations, escalation, and human approval.
Microsoft Foundry · Agent Orchestration · RAG · Observability
GitHub Repo: -
- Full-Stack Software Projects**
Full-Stack Engineering · Application Architecture
React · Vite · Node.js · Express · MongoDB · PostgreSQL · Prisma
A portfolio of end-to-end full-stack systems demonstrating frontend architecture, backend service development, REST API design, database architecture, authentication and authorization, state management, asynchronous application flows, component architecture, and integration between client applications, backend services, databases, and external APIs.
Projects span document and relational data models, JWT-based authentication, protected API workflows, Mongoose and Prisma data access, PostgreSQL schema and SQL operations, third-party API integration, external service integration, AI and LLM integration, retrieval workflows, validation, error handling, and application lifecycle management across independently structured frontend and backend systems.
GitHub Repo: [github.com/keenosmith-del/personal-productivity-desktop](#)
GitHub Repo: [github.com/keenosmith-del/ai-entity](#)
GitHub Repo: [github.com/keenosmith-del/music-api](#)
GitHub Repo: [github.com/keenosmith-del/enterprise-workspace](#)

Portfolio: [keenosmith.vercel.app](#)

06 Skills

GitHub: [github.com/keenosmith-del](#)
CodeWars: [codewars.com/users/keenosmith-del](#)
LeetCode: [leetcode.com/u/keenosmith/](#)
freeCodeCamp: [freecodecamp.org/keenosmith](#)

Languages

JavaScript · Python · Java · SQL · HTML · CSS

Frontend

React · Vite · React Router · Responsive Web Development · UI Architecture · Component Architecture

Backend

Node.js · Express · REST APIs · Authentication · JWT · Backend Architecture

Databases

MongoDB · Mongoose · PostgreSQL · Prisma · Database Design · Relational Data Modelling

AI / Machine Learning

LLM Integration · RAG · Prompt Engineering · Agentic AI · PyTorch · Neural Networks · Transformers

Cloud / DevOps

AWS · Git · GitHub · CI/CD · Docker · Kubernetes · Cloud Deployment

Engineering

Full-Stack Development · Software Architecture · API Design · State Management · Testing & Debugging · Version Control